

Rahul Rawat

DATA SCIENTIST

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PROFILE

Data Science and Analytics Master student at SRH Heidelberg based in Mannheim, with hands on experience building forecasting and market analysis pipelines on real environmental and financial data. I have shipped a recursive time series forecasting pipeline with anti leakage rules and prediction intervals at eRay GmbH, a fairness aware credit scoring system with a full EU regulatory write up covering the EU AI Act and GDPR, and a Random Forest driven study on the economic impact of global climate events. Comfortable in Python, GitHub, and cloud data workflows, I am the right fit for advancing environmental market modeling for the EU Emission Trading System.

EXPERIENCE

eRay GmbH

Oct 2025 to Mar 2026

Data Scientist

- Built an end to end recursive time series pipeline forecasting chlorophyll a, turbidity, pH, and dissolved oxygen for a German lake, delivered as a six month collaboration with SRH University Heidelberg.
- Benchmarked six models head to head, Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, and used CatBoost multi quantile regression to produce asymmetric 80 percent prediction intervals for decision support under uncertainty.
- Enforced strict anti leakage rules across the pipeline, surfacing the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not without live optical sensors.
- Reconstructed missing winter readings with MICE imputation and engineered a synthetic winter decay forecast canvas so tree based models stopped flatlining during recursive prediction on a limited prediction horizon.
- Wrapped the whole pipeline in an orchestrator with gate checks and velocity and ecological bounds that halts on failed imputation rather than letting bad data cascade downstream.

EDUCATION

M.Sc. Data Science and Analytics

Apr 2025 to Present

SRH University of Applied Sciences Heidelberg

Bachelor of Technology in Computer Science

2019 to 2023

GL Bajaj Institute of Technology and Management

PERSONAL PROJECTS

Economic Impact Analysis of Global Climate Events

Python with Pandas and scikit learn

Random Forest

statistical modelling

Matplotlib and Seaborn

- Executed an end to end data science project transforming raw global event datasets into structured insights for resource allocation and risk assessment, close in spirit to how environmental market signals feed EU ETS decisions.
- Performed advanced data preparation and cleansing across outliers, imputation, and normalisation to build a high quality data foundation before any modelling.
- Developed Random Forest models to analyse correlations between event duration and financial impact, extracting clear business relevant insights on where economic risk concentrates.
- Communicated complex statistical findings to non technical stakeholders through comprehensive visual reports and calibrated confidence statements.

CreditIQ, Fairness by Design Credit Scoring System

Python

scikit learn

AIF360

SHAP

Streamlit

- Lifted the model's Disparate Impact ratio from a failing 0.79 to a compliant 0.88, clearing the EU AI Act and AGG 80 percent fairness threshold, an exercise in defensible modelling under EU regulation.
- Diagnosed a hidden intersectional bias where younger women were still being penalised even after single axis age bias was fixed, using SHAP, then designed a four way age by gender threshold matrix to correct it without over correcting into reverse discrimination.
- Cut the false negative rate from 44 percent to 16.7 percent while holding accuracy at 75 percent, documenting the fairness accuracy trade off as a deliberate and regulator defensible decision.
- Shipped a Streamlit decision support tool giving finance managers a recommendation plus a plain language LLM generated explanation, keeping a human in the loop per GDPR Article 22 and EU AI Act Article 14.
- Backed the pipeline with unit tests at 100 percent branch coverage and a full regulatory write up covering EU AI Act Annex III, GDPR, a model card, and attack vectors.

Movie Analytics and ML Pipeline, Cloud Native Data Platform

GCP BigQuery

Cloud Run

GCS

Cloud Scheduler

BigQuery ML

Python

SQL

Looker Studio

- Built an end to end batch pipeline pulling data from a public API into a GCS data lake, processed through Bronze to Silver to Gold medallion architecture in BigQuery, fully automated on a Cloud Scheduler trigger with no manual steps, the same shape of pipeline environmental market analysts rely on.
- Engineered the Silver layer for trustworthy analytics with schema enforcement, safe type casting, deduplication via window functions, and normalisation into a relational model.
- Trained a BigQuery ML classifier as a downstream prediction layer, deliberately splitting features into two tables to prevent data leakage and keeping only pre event signals.
- Built Gold layer aggregates and a five page Looker Studio dashboard answering concrete business questions, plus an ML early warning view, and secured the system with a least privilege service account and Secret Manager.

RESEARCH AND THESIS

Bachelor Thesis, Diabetes Prediction Using Machine Learning

GL Bajaj Institute of Technology and Management, IEEE style paper

- Built a full end to end machine learning pipeline comparing six classifiers on a clinical dataset of 768 patients, using 10 fold cross validation and per model confusion matrices.
- Caught biologically impossible zero values that the original authors had overlooked, then applied IQR based outlier removal and proper imputation to restore data integrity.
- Chose ROC AUC over accuracy as the headline metric for a 65 to 35 imbalanced dataset, avoiding a misleadingly rosy accuracy figure that would have hidden real error patterns.
- Wrote up findings as an IEEE style paper including an honest section on limitations and what to do differently in a follow up study.

CERTIFICATIONS

- AWS Academy Graduate: AWS Academy Cloud Foundations, issued July 2025.
- SAS Certified Specialist: Visual Business Analytics Using SAS Viya, issued May 2025.
- Google Data Analytics: Foundations, Data, Data, Everywhere, issued April 2025 on Coursera.

ACHIEVEMENTS

- USAII Global AI Hackathon 2026: Finalist at Graduate Level, event 14 to 21 June 2026, awarded by the United States Artificial Intelligence Institute for innovation, technical creativity, and applied AI on real world challenges. Hackathon ID 830382652.

LANGUAGES

English: fluent, professional working proficiency. **German:** B1, in progress.